

Chemist: KR Completed Date: 1/29/26Checker: TLR Review Date: 2/25/26IMIS - 1460Analyte(s) - Hydrogen Fluoride (as F⁻)**Sample Type (choose one):**☒ **QCSM**QC sample numbers: 131268 - 131279**Data Package Documents & Records**

- ☒ A) Initial Standard Screen Print
☒ B) Intermediate Standard Screen Print
☒ C) Batch Preparation Screen Print
☒ D) Batch Preparation Volumes Screen Print
☒ E) QCSM Prep Worksheet - Input (each analyte)
☒ F) QCSM Prep Worksheet - Spiking (each analyte)
☒ G) Certificate of Analysis (for each stock)

☐ **Internal Proficiency/Demonstration of Competence****Data Package Documents & Records**

- ☐ A) Initial Standard Screen Print
☐ B) Intermediate Standard Screen Print
☐ C) IP Prep Worksheet - Input
☐ D) IP Prep Worksheet - Spiking
☐ E) Certificate of Analysis (for each stock)
☐ F) 91A(s)

☐ **State Exchange: Round****Data Package Documents & Records**

- ☐ A) Initial Standard Screen Print
☐ B) Intermediate Standard Screen Print
☐ C) Prep Worksheet - Input
☐ D) Prep Worksheet - Spiking
☐ E) List of Participating Labs
☐ F) Letter Sent to Labs/Reporting Sheet
☐ G) Certificate of Analysis (for each stock)
☐ H) 91A(s)

Other support documents/records (specify):

☐ Z1) _____**Standard Documentation**

- | ✓A | ✓C | |
|-------------------------------------|-------------------------------------|--|
| ☒ | ☒ | 1) Density/Concentration of spiking material |
| ☒ | ☒ | 2) Purity |
| ☒ | ☒ | 3) Manufacturer |
| ☒ | ☒ | 4) Lot Number |
| ☒ | ☒ | 5) Standard ID (if necessary) |
| ☒ | ☒ | 6) Barcode |
| ☒ | ☒ | 7) Date Received/Prepped/Expires |

Media Documentation

- | | | |
|-------------------------------------|-------------------------------------|---------------------|
| ☒ | ☒ | 8) Media Type |
| ☒ | ☒ | 9) Manufacturer |
| ☒ | ☒ | 10) Catalog # |
| ☒ | ☒ | 11) Lot # |
| ☒ | ☒ | 12) Expiration Date |

Sample Labeling

- | | |
|-------------------------------------|--|
| ☒ | 13) Labels attached to sample and packaged/stored correctly |
| ☒ | 14) Sample number on each label matches correct sample number on preparation worksheets |
| ☒ | 15) QCSM Only. Labels attached to outside of box/container identify analyte, IMIS, media |

Documentation Review

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☒ | 16) All hand written entries in indelible ink; corrections use single strikeout (dated & initialed) |
| ☒ | ☒ | 17) Analyst's initials & date on top sheet of associated, or linked, pages of documents |
| ☒ | ☒ | 18) Data exclusion is adequately justified and documented |
| ☒ | ☒ | 19) Documentation produced is sufficient to allow complete re-construction of sample preparation events |

Reagent & Standard Usage

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☒ | 20) Reagents and standards used (spiking solutions & stocks) are <u>not</u> expired |
| ☒ | ☒ | 21) Standard and reagent usage is documented in an <i>explicitly traceable</i> manner (Q-008) |

Data Management Review

- | | | |
|-------------------------------------|-------------------------------------|-----------------------------------|
| ☒ | ☒ | 22) Calculations verified correct |
|-------------------------------------|-------------------------------------|-----------------------------------|

Comments: _____

AF

KR 1/29/26

Initial Standard

Manufacturer

Aldrich Chemical Company [Sigma]

Status

In-Use

Lot No

MKCK9071

Search

Bar Code No

STD02190-0001-

Search

% Purity

99.1

Date Received

02-JUN-2021

Comments

F/NaF = 19/42 = 0.4524

Date

Reevaluated

Date Expires

02-JUN-2026

Physical State

Solution

Manuf Titration

SLTC Titration

Analyte Composition

IMIS	Analyte Name	CAS	Part/Density	Unit	Concentration
1460	Hydrogen Fluoride (as F)	7664-39-3	4524	mg/mL	.4483284

Save



Ⓟ
KR 1129126

Intermediate Standard

Standard Name	Preparer's Name	First Name	Initial	Update
1460-20260129-KR-1	Ramji	Karpagavalli	KR	
Prepared Date	Expiration Date	Disposition	State	Re-evaluation Date
29-JAN-2026	29-JAN-2027	In-Use	Solution	
Comment	Made with DI water.			

Parent Standards

MKCK9071

Analyte Composition

IMIS	Analyte Name	Amount	Final Volume	Concentration	Unit
1460	Hydrogen Fluoride (as F)	302.54	15	9.041912	mg/mL

QC Batch Preparation - QC DIVISION - 20110317

CR 1/29/26

QC Batch

Last Name	Ramji	Media Lot Number	MD039407
Media Name	MCEF, BUP Na2CO3	Initials	KR
Media Short Desc	MCEF, 37-mm, BUP, Na2CO3; SKC 225-9001	Media Expiration Date	31-MAY-2027
Media Desc		Comments	
Prepared On	29-JAN-2026	Comments Analyst	
Expiration Interval	12		
Expiration Date	29-JAN-2027		

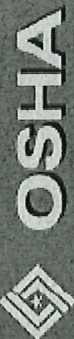
Initial Std Source Intermediate Std Source

Intermediate Std Source

Name	1460-20260129-KR-1

QC Sets

Lab Number	131268	Spknt Type	Gilson 10-100uL	Serial No	UH14270
Last QC Sample Number	131267	How Many Qc Sets	3	Samples per Set	4



To Sol Vol

(E)

KR 1/29/26

SLTC QCSM Preparation Worksheet**Input**

Date: 1/29/2026
of sets to spike: 3

First QC #: 131268

Amounts Spiked (µL)

Spike 1: 10
Spike 2: 20
Spike 3: 40
Blank: 0

Analyte

Name: HF (as F-)
IMIS: 1460
Synonym: Hydrogen Fluoride
CAS: 7664-39-3

PEL(mg/m³): 2
Rec. Air Volume(L): 90

Standard Stock

Supplier: Sigma Aldrich
Concentration/Density of spiking material: 9.041912 µg/µL
Purity: 1.000
Molecular Weight: 18.998
Lot #: MKCK9071
Standard ID: 1460-20260129-KR
Bar Code: STD02190-0001-001
Date Rec'd/Prep'd: 1/29/2026
Date Expires: 1/29/2027

Media

Media Type: MCEF, BUP Na₂CO₃
Manufacturer: SKC
Catalog #: 225-9001
Lot #: MO039407
Expiration Date: 5/31/27
Date Prepared: 1/29/2026
Date Expires: 1/29/2027
Prepared by: KR
Target Analytical Technique(s): SIE

General comments: Spiked using Gilson Pipetman 10-100uL, BC: UH14270.

KR 1/29/26

[illegible]

302325

K2129126

SIGMA-ALDRICH®

sigma-aldrich.com

3050 Spruce Street, Saint Louis, MO 63103, USA

Website: www.sigmaaldrich.comEmail USA: techserv@sial.comOutside USA: eurtechserv@sial.com

Certificate of Analysis

Product Name:

Sodium fluoride – 99.99% trace metals basis

Product Number: 215309
Batch Number: MKCK9071
Brand: ALDRICH
CAS Number: 7681-49-4
MDL Number: MFCD00003524
Formula: FNa
Formula Weight: 41.99 g/mol
Quality Release Date: 16 SEP 2019

NaF

Test	Specification	Result
Appearance (Color)	White	White
Appearance (Form)	Powder	Powder
X-Ray Diffraction	Conforms to Structure	Conforms
Purity (Titration by NaOH)	≥ 98.5 %	99.1 %
Purity	Meets Requirements	Meets Requirements
99.99% Based on Trace Metals Analysis		
Trace Metal Analysis	≤ 150.0 ppm	103.8 ppm
Aluminum (Al)		4.7 ppm
Boron (B)		2.4 ppm
Barium (Ba)		0.3 ppm
Calcium (Ca)		48.0 ppm
Chromium (Cr)		0.7 ppm
Iron (Fe)		11.0 ppm
Magnesium (Mg)		2.9 ppm
Manganese (Mn)		0.8 ppm
Nickel (Ni)		0.6 ppm
Strontium (Sr)		0.3 ppm
Titanium (Ti)		0.3 ppm
Zinc (Zn)		0.3 ppm
Cesium (Cs)		7.2 ppm
Potassium (K)		2.5 ppm
Lithium (Li)		0.4 ppm
Rubidium (Rb)		0.6 ppm

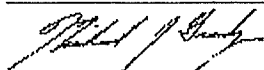
Sigma-Aldrich warrants, that at the time of the quality release or subsequent retest date this product conformed to the information contained in this publication. The current Specification sheet may be available at Sigma-Aldrich.com. For further inquiries, please contact Technical Service. Purchaser must determine the suitability of the product for its particular use. See reverse side of invoice or packing slip for additional terms and conditions of sale.

3050 Spruce Street, Saint Louis, MO 63103, USA

Website: www.sigmaaldrich.comEmail USA: techserv@sial.comOutside USA: eurtechserv@sial.com

Certificate of Analysis

Product Number: 215309
Batch Number: MKCK9071



Michael Grady, Manager
Quality Control
Milwaukee, WI US

Sigma-Aldrich warrants, that at the time of the quality release or subsequent retest date this product conformed to the information contained in this publication. The current Specification sheet may be available at Sigma-Aldrich.com. For further inquiries, please contact Technical Service. Purchaser must determine the suitability of the product for its particular use. See reverse side of invoice or packing slip for additional terms and conditions of sale.